

PlantPulse — Incident Report

PlantPulse Demo Site — Process Unit 7 · Generated 2026-07-22 19:20

Incident	Overpressure Risk While the Relief Path Is Degraded
Zone	Z1
Window	2026-07-22T16:00:00 → 2026-07-22T18:00:00
Ground-truth severity	CRITICAL
Single-channel alarm would fire?	No — this is exactly the compound pattern PlantPulse exists to catch

Compound Risk at Peak

83 / 100

Confidence: 47%

Contributing Factors

- Sensor: +55%
- Permit: +25%
- Maintenance: +10%
- Equipment: +6%
- Weather: +4%

Root Cause (Knowledge Graph)

TANK-01 shows Pressure rising, Tank Temperature rising; VALVE-01 shows Relief Capacity falling; MT-02 (Relief valve function test) is in progress; a hot-work permit is active in the same restricted zone.

SOP Response — SOP-HOTWORK-01: Hot Work Permit Suspension (Ignition Risk During Process Excursion)

Source: OISD-STD-105 (Work Permit System) — hot work permit provisions; Factories Act 1948, Section 38 (Precautions in case of fire)

- 1 Immediately suspend the active hot-work permit and stop all spark- or flame-producing work
- 2 Remove hot-work equipment and other ignition sources from the zone
- 3 Verify fire watch personnel and extinguishing equipment are stationed before any resumption is considered
- 4 Do not reissue the permit until the pressure/temperature excursion is confirmed resolved and relief capacity is restored to 100%
- 5 Re-verify the gas-free / LEL certificate before hot work resumes
- 6 Document the near-miss and route the permit through the issuing authority for re-approval

Estimated risk reduction if followed: 7% (peak compound risk 83 → 77, computed by re-scoring a mitigated counterfactual through the same fusion model)

Detection Performance

Across 20 independently seeded evaluation days, PlantPulse detected this archetype in **100%** of instances with a median lead time of **38 minutes** over an extrapolated conventional single-sensor alarm. Source: engine/metrics/results.json.